CV

AI CAD-to-Cost Analysis

CostVision · Powered by Claude AI · Manufacturing Intelligence Platform

AI-POWERED CAD ANALYSIS REPORT

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Geometry to Process to Cost · Fully AI-reasoned

18

PRCR002 – complex machined/cast housing (mis-classified as sheet metal by Stag

277 × 223 × 182 mm · 1037.1 cm³ · Al 2.800 kg / Steel 8.141 kg

Manufacturability: 18/100 · Confidence: Low

§1 — Geometry & Part Summary

Bounding Box	**276.8 × 223.2 × 181.9 mm**	Surface Area	**1370.1 cm²**
Volume	**1037.11 cm³**	Weight (Al)	**2.800 kg**
Weight (Steel)	**8.141 kg**	Weight (Plastic)	**1.089 kg**

§2 — Detected Features

Feature Type	Count	Significance	Description
Non-uniform wall thickness (0.45–169.95mm, mean 34.1mm)	1	High	Wall thickness varies by two orders of magnitude — incompatible with constant-gauge sheet metal forming
Cylindrical/Conical/Toroidal faces	221	High	132 cylinders + 55 cones + 34 tori — turned/cast/machined bosses, fillets and shafts, not stamped features
Free-form B-spline surfaces	76	High	76 organic surfaces — not producible by bend/pierce tooling
Undercuts	44	High	44 undercut faces — require side-action tooling or multi-axis machining, impossible in a simple progressive die
Drilled/bored holes	21	Medium	21 holes r<30mm, several with threads
Boss/shaft features	10	Medium	10 boss/shaft features up to r=57mm — thick solid sections

§3 — Material Analysis

AI-suggested

Primary Material: **DC01 mild steel (sheet-metal framework material choice) (45% confidence)**

No aluminium signal present (no lightweighting note, no thin uniform 2-4mm HPDC wall). Geometry shows thick, dense boss/shaft/threaded features typical of a mechanical/structural component, so per guidance we default to steel rather than aluminium. However, the wall-thickness spread (0.45–170mm) and face mix (cylinders/cones/tori/B-splines, 44 undercuts) is fundamentally inconsistent with sheet metal — this is almost certainly a cast-and-machined or forged steel part, not a stamping. DC01 is used here only to satisfy the forced sheet-metal cost framework.

Alternatives: **1045 medium carbon steel (bar/forging) (30%) · SG iron (ductile cast iron) - if cast (15%) · Aluminium 6061 (unlikely, no signal) (10%)**

§4 — Process Recommendations

Process	Commodity	Confidence	Cycle Time (hr)	Reasoning
Sheet metal progressive stamping (as instructed by Stage-1)	sheet_metal	20%	0.0200	Stage-1 classifier flagged sheet_metal, but geometry (non-uniform 0.45–170mm walls, 132 cylinders, 55 cones, 34 tori, 76 free-form faces, 44 undercuts, threaded bosses) is not achievable by blank/bend/pierce tooling. Retained only because required by task framework.
Investment or sand casting + machining	cast_and_machine	65%	0.4500	High free-form content, undercuts, cored bosses and variable wall sections strongly indicate a cast (steel/ductile iron) part finished by CNC drilling/threading — a far better geometric fit than stamping.
5-axis CNC machining from billet	machining	40%	1.0680	Bottom-up CNC estimate (1.068hr, 3 setups) already computed and is a viable fallback if low-volume prototyping is required before tooling investment.

§5 — Manufacturability Risks (Score: 18/100)

Severity	Feature / Area	Description	Recommended Action
High	**Process mismatch**	Part geometry (wall thickness Ã= cylinders/cones/tori, 44 undercuts) is incompatible with sheet-metal stamping despite Stage-1 classification	Re-classify as casting or machining before committing to progressive-die tooling
High	**Undercuts (44)**	Cannot be formed by straight punch/die motion; would require multiple secondary hits or complete re-tooling as cast part	Move to casting with side cores, or split into multiple sheet-metal sub-assemblies
Medium	**Minimum wall 0.45mm**	Extremely thin local section is at risk of tearing during piercing/forming	Increase local wall to "e1.0mm or mov to a separate thinner insert
Medium	**Threaded features**	Threads cannot be formed in-die; require secondary tapping/thread-forming operation not covered by the stamping cycle	Add post-forming tapping station or use self-clinching/flow-form inserts

§6 — DFM Issues (sheet_metal)

Severity	Area	Description	Impact	Fix
Critical	**44 undercuts**	Undercut faces cannot be released from a straight-motion progressive die; would require side-action tooling or a full process change to casting/machining.		
Critical	**Non-uniform wall thickness (0.45–169.95mm)**	Two-order-of-magnitude wall variation is incompatible with constant-gauge sheet forming and will cause tearing, buckling or non-fill in a stamped blank.		
High	**76 free-form B-spline surfaces**	Organic surfaces cannot be produced by bend/pierce tooling and imply a cast or machined origin.		

Severity	Area	Description	Impact	Fix
High	**221 cylindrical/conical /toroidal faces**	High count of turned/cast features (bosses, fillets, shafts) indicates a rotationally-machined or cast part, not a stamping.		
Medium	**21 drilled/threaded holes**	Threads and bored holes require secondary machining operations not part of the stamping cycle.		

§7 — Cost Range & Suggested Inputs

OPTIMISTIC	MOST LIKELY	CONSERVATIVE
£0.00	**£0.00**	**£0.00**

Net Weight	**8.141 kg**	Material	**mat-dc01**
Recommended Process	**sheet_metal**	Cycle Time	**1.0680 hr/part**
Setup Time	**0.750 hr**	Operations	**3 ops**
Operations Detail	**Blank (mach-haas-vf2, 0.0006 hr) Progressive bend/pierce (25 bends + pierce) (mach-haas-vf2, 0.0006 hr) Secondary drilling/tapping (21 holes + threads) (mach-drill, 0.1750 hr)**		

Process-Specific Parameters

Casting Subtype	**sand**
Die/Mould Cost	**£33,000**
Die Life	**100,000 shots**
Cavities	**1**
Yield	**65.0%**
Flash Weight	**0.800 kg**
Yield	**75.0%**
Die Cost	**£180,000**
Strokes	**8**
Cavities	**1**
Wall Thickness	**34.1 mm**
Mould Cost	**£200,000**
Mould Life	**500,000 shots**

Projected Area	**615.0 cm²**

§8 — AI Analysis Explanation